

CONTAINER: Bounding Credential-Compromise Impact in IEEE 2030.5 DER Systems

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Abstract—The IEEE 2030.5 standard governs communication with distributed energy resources (DERs), but its trust model binds a device’s long-lived bootstrap identity to its operational authorization and forbids online certificate revocation, so a stolen credential retains control authority for a lifetime the standard permits to be indefinite. We argue that the harm of such a compromise is not captured by static reachability, and introduce a four-dimension model separating cyber reach, state-dependent commandable flexibility, retained authority, and feeder consequence; a worked counterexample gives two compromises with identical reach yet different flexibility, persistence, and feeder outcome. Guided by the model, CONTAINER replaces the conflated device credential with attested, short-lived, scope-bound operational credentials, using non-renewal rather than revocation for containment, with incremental deployment through native and gateway-proxy tiers. We build a working credential service with real X.509 and mutual TLS and a real feeder power-flow pipeline, and report a preliminary evaluation on the IEEE 8500-node feeder: ten security invariants hold, per-command enforcement bounds authority at credential expiry, attested issuance costs about 0.2 ms, and narrowing authorization keeps a compromise’s induced overvoltage within limits where a legacy credential drives it far outside. These reduced-scale results precede the full pre-registered evaluation.

Index Terms—Critical infrastructure protection, distributed energy resources, IEEE 2030.5, remote attestation, authorization, cyber-physical security.

I. INTRODUCTION

Distributed energy resources now participate in grid operations at a scale that makes their control interface a systemic concern. IEEE 2030.5 (Smart Energy Profile 2.0), profiled for distribution use by the Common Smart Inverter Profile (CSIP), is the application protocol through which utilities and aggregators dispatch inverter-based resources. An adversary who obtains a credentialed identity in this ecosystem does not merely read data; it can command real and reactive power on physical hardware connected to the grid.

Two properties of the IEEE 2030.5 trust model make this worse than a conventional credential-theft problem. First, the device’s bootstrap identity and its operational authorization are the same long-lived certificate: there is no separate, short-lived operational credential to lose. Second, the standard forbids certificate revocation; it directs that certificate authorities not maintain revocation lists or run status responders, and that clients and servers not check them [1]. A stolen credential is therefore valid until it expires, and its lifetime may be indefinite.

A natural objection is that this is a solved problem: model the DER authorization surface as a privilege graph and measure blast radius as reachability, as a mature line of work does for enterprise networks [2]–[4]. We argue this reduction fails. What determines the damage of a DER compromise is not only which nodes an adversary can reach, but how much physical capacity those nodes command, how long the adversary retains that capacity, and what the feeder does in response under the grid state at the time. Reachability over a static privilege graph is a function of the attack locus and the graph alone; the quantities that matter are functions of the defensive policy and of exogenous physical state, and neither is topological.

Contributions

- A *four-dimensional impact model* (BR_{reach} , BR_{flex} , BR_{auth} , BR_{phys}) with explicit domains, units, and computation procedures, separating cyber reach from state-dependent commandable flexibility, retained authority, and feeder-level consequence.
- A set of *separation propositions* showing that reachability-only measurement omits scope, time, and feeder state, with a worked counterexample of two compromises that share topological reach but diverge in flexibility, persistence, and feeder outcome.
- A *threat model* that separates bootstrap identity, operational credential, and authorization scope, and names the baseline standard’s conflation of the first two as the driver of unbounded persistence.
- *CONTAINER*, an incrementally deployable redesign using attested, short-lived, scope-bound operational credentials with non-renewal as the containment mechanism.
- An implemented cyber-side analysis engine that computes the four dimensions for a compromise locus under a parameterized authorization policy, a working credential service using real X.509 and mutual TLS, and a preliminary real evaluation on the IEEE 8500-node feeder under a pre-registered protocol.

We state scope plainly. The formal model, the separation propositions, and the cyber-side engine are complete and tested. The credential service and the feeder pipeline are implemented, and we report a preliminary evaluation on the IEEE 8500-node feeder, confirmed on the IEEE 123-bus feeder: the results are measured but at reduced scale, two public feeders, one solver, up to ten seeds, on a workstation.

The full pre-registered evaluation, with a cross-solver check, constrained hardware, and the full seed budget, is future work.

II. BACKGROUND AND THREAT MODEL

A. Authorization in IEEE 2030.5 / CSIP

IEEE 2030.5 uses mutual TLS with X.509 device certificates. A device's identity is the certificate-derived LFDI/SFDI, and its authorization is scoped by a `FunctionSetAssignments` resource that binds it to one or more `DERPrograms`. Physical control authority is carried by the `DERControlBase` control modes: active-power capping (`opModMaxLimW`), signed fixed active power for storage charge and discharge (`opModFixedW`), reactive power (`opModFixedVar`, `opModVoltVar`), volt-watt and frequency-watt curves, and the hard connect and energize controls [5]. The standard specifies certificate-based authentication and function-set scoping; it does not mandate a single canonical access-control object.

B. Certificate validity and revocation

The trust model has two properties central to this work [1]. Device certificates have indefinite lifetime, and revocation is prohibited: certificate authorities are directed not to maintain revocation lists or run status responders, and endpoints are directed not to check them. The stated reasons against online revocation are concrete design constraints, not incidental: field devices have no guaranteed connectivity, a hard-fail check destroys availability while a soft-fail check is trivially bypassed by blocking the responder, revocation presupposes a secure time source that inverters may lack, and no party is defined as responsible for reporting a compromised device. The sanctioned substitute is a local allow or deny list. The consequence for our threat model is direct: once a credential is compromised, there is no online mechanism to withdraw it, so its authority persists until it expires.

C. Three separated layers

We separate three layers that the baseline standard conflates. *L1, bootstrap identity*: the hardware-rooted device certificate. *L2, operational credential*: the credential that authorizes live sessions to an aggregator or DER management system. *L3, authorization scope*: the server-side binding from an identity to the function sets it may invoke. In baseline IEEE 2030.5, L1 and L2 are the same long-lived certificate, with no attestation gate and no revocation path. This conflation, not any single cryptographic weakness, is what makes a compromise persist for the certificate lifetime.

D. Compromise loci and adversary

We consider four compromise loci of increasing consequence: a single device, an edge gateway, an aggregator or DER management system, and a utility-side issuing role. The adversary obtains a valid credential at one locus (by key extraction, session theft, or a misconfigured or compromised server granting scope) and issues protocol-conformant control

TABLE I
STANDARDS CLAIMS AND VERIFICATION STATUS.

ID	Claim	Status
S1	Device certificate may be long-lived/indefinite	pending 2023 clause
S2	CRL use prohibited/unsupported	SAND §4.1; confirm 2023
S3	OCSP prohibited/unsupported	SAND §4.1; confirm 2023
S4	Local allow/deny listing available	SAND §4.1; confirm 2023
S5	Certificate replacement/re-enrollment possible	pending 2023 clause
S6	LFDI/SFDI derives from certificate	pending 2023 clause
S7	FunctionSetAssignments scopes operations	pending 2023 clause
S8	Scheduled DER controls can persist	pending 2023 clause

commands within that credential's authorization. The adversary does not break TLS or the underlying cryptography. The utility-side issuing role is the ceiling on what any credential discipline can bound, because it can subvert issuance itself; we treat it as the boundary of the design's protection rather than a case the design defeats.

E. Credential lifecycle and existing containment

The motivation depends on exact certificate-management behavior, which differs across editions: IEEE 2030.5-2018 [6], IEEE 2030.5-2023 [7] (which supersedes it, published December 2024), and the profile in force [8]. We separate four questions the literature tends to conflate: whether a certificate can be cryptographically revoked; whether a server can locally deny an identity despite a valid certificate; whether a device can be re-enrolled or re-keyed; and whether an active session or an already-issued control remains effective after credential expiry. Lack of CRL/OCSP bounds only the first. Table I lists the claims our argument uses and their verification status; the revocation rows are sourced to the Sandia analysis, and the clause-level rows are marked pending direct verification against the applicable edition [1].

F. Assets, adversary, and goals

Protected assets: DER control integrity, fleet authorization, operational and attestation keys, and feeder stability and service continuity. *Adversary capabilities*: steals an operational key (and, where tested, a bootstrap key), compromises firmware, steals an active session, compromises a gateway or aggregator, issues protocol-conformant commands, and observes authorized telemetry. *Exclusions*: breaking standard cryptography, physical destruction, and compromising both the root issuer and all audit controls, which we treat as the protection boundary rather than a defended case. *Security goals*: least authority, bounded new-session authority, bounded active-session authority, bounded command-effect duration, fresh-key containment, replay resistance, safe degradation, and auditability. *Non-goals*: defending against malicious root issuance without added governance, detecting every hardware

attack, replacing grid protection, and proving safety for every feeder.

III. THE FOUR-DIMENSIONAL AUTHORIZATION-COMPROMISE IMPACT MODEL

We model the impact of an authorization compromise along four dimensions rather than a single scalar radius, because the four quantities have different mathematical types: a set, a state-dependent power region, a set of durations, and a feeder-physics functional. We keep the BR notation for continuity but describe each as a dimension.

a) Substrate.: A static privilege graph $G = (V, E)$ has principals and resources as nodes and access relations as edges; it is a snapshot with no time axis and no physical state. A compromise $x \subseteq V$ is a set of compromised credentials. A protection policy $\pi = (L, \text{Att}, A)$ encodes a credential-lifetime function L , an attestation-gated renewal predicate Att , and an authorization scope map A . The device and grid operating state at time t is $u_t \in U$, and F is the feeder power-flow and protection model; neither u_t nor F appears in G .

b) Cyber reach (BR_{reach}).: The set of principals, resources, or DER endpoints to which the compromised authority can submit an authenticated request, $\text{ReachSet}(x, \pi)$, the least fixed point of the π -enabled access relation from x . We report reachable principals, reachable DERs, reachable sites, the fraction of fleet, and the authorization paths by compromise locus. Reach is a cyber quantity: reaching a server is not the same as controlling a physical resource, and we keep the two separate.

c) Commandable flexibility (BR_{flex}).: Capacity is state dependent, so we define it through a feasible command set rather than nameplate. For DER d at time t ,

$$\mathcal{C}_d(t; u_t, A_\pi(d)) \quad (1)$$

is the set of set-points an adversary holding scope $A_\pi(d)$ can command given the device and grid state u_t : present set-point, state of charge, irradiance, inverter apparent-power and ramp limits, and availability. The aggregate reachable flexibility over the reachable DERs is the achievable active- and reactive-power intervals $[\Delta P^-, \Delta P^+]$ and $[\Delta Q^-, \Delta Q^+]$. A conservative scalar for tables is

$$\text{BR}_{\text{flex}}(t) = w_P(\Delta P^+ - \Delta P^-) + w_Q(\Delta Q^+ - \Delta Q^-), \quad (2)$$

with weights w_P, w_Q stated wherever the scalar is used; the implementation retains the full vector and region. Read-only or empty scope yields the singleton current set-point, hence zero flexibility.

d) Retained authority (BR_{auth}) and exposure (BR_{exp}).: Certificate lifetime alone understates persistence, so we separate three components: T_{cred} , the time until the stolen operational credential is no longer accepted for a new session; T_{sess} , the time until sessions authenticated with it can no longer issue commands; and T_{cmd} , the time until queued, scheduled, or latched controls cease to affect the DER. Effective retained authority is

$$\text{BR}_{\text{auth}} = \max(T_{\text{cred}}, T_{\text{sess}}, T_{\text{cmd}}), \quad (3)$$

TABLE II
THE FOUR IMPACT DIMENSIONS.

Dim.	Type	Unit	Computation
BR_{reach}	set	count	fixed point of π -enabled access from x
BR_{flex}	interval	kW, kVAr	$\sum_d \mathcal{C}_d(t; u_t, A_\pi(d))$ over reachable DERs
BR_{auth}	duration	s	$\max(T_{\text{cred}}, T_{\text{sess}}, T_{\text{cmd}})$
BR_{phys}	scalar	node-min, p.u.	$\max_a J_V(F, u, a)$ over the feeder model

and the more informative capacity-time exposure is

$$\text{BR}_{\text{exp}} = \int_{t_0}^{t_{\text{end}}} \text{BR}_{\text{flex}}(t) \mathbf{1}[\text{adversarial authority effective at } t] dt, \quad (4)$$

in kW·h or kVAr·h. This distinction matters: certificate expiry does not by itself end an open session or an already-issued long-duration control, so a design that shortens T_{cred} without enforcing T_{sess} and T_{cmd} does not bound exposure.

e) Physical consequence (BR_{phys}).: Physical consequence is the worst outcome an adversary can drive within its authority over a horizon H . For an attacker action trajectory $a_{0:H} \in \mathcal{A}(x, \pi, u_{0:H})$ constrained to authorized functions and device limits,

$$\text{BR}_{\text{phys}} = \max_{a_{0:H} \in \mathcal{A}} J(F, u_{0:H}, a_{0:H}), \quad (5)$$

where J is a declared impact objective. Our primary objective is the voltage-violation area, the node-time outside the allowed band weighted by magnitude: for node n with voltage $V_n(t)$ and bounds $V_{\min}, V_{\max}$,

$$J_V = \sum_{n,t} \left[\max(0, V_n(t) - V_{\max}) + \max(0, V_{\min} - V_n(t)) \right] \Delta t. \quad (6)$$

Secondary metrics are maximum per-unit deviation, violating-node count, protection trips, regulator tap operations, energy not served, unnecessary curtailment, and recovery time. We report both a worst-case optimizing attacker and an operationally realistic attacker that uses only telemetry available through the compromised authority, so the paper does not report only an omniscient upper bound.

Table II summarizes the four dimensions, their mathematical types, units, and computation. The distinct types are the reason we present them as dimensions rather than forcing a single scalar radius.

IV. WHY REACHABILITY-ONLY MEASUREMENT IS INSUFFICIENT

We do not claim that no graph model can represent time or physical impact; temporal, probabilistic, and attributed cyber-physical attack-graph models exist and can encode some of these quantities. Our claim is narrower and, we argue, more defensible:

Static, unweighted reachability is insufficient to compare DER credential-containment policies, because it omits authorization scope, temporal valid-

TABLE III
IMPACT DIMENSIONS VERSUS REACHABILITY-ONLY METRICS.

Dimension	Closest metric	What reachability-only omits
BR_{reach}	logical reachability [2]	(overlap; the cyber-reach dimension)
BR_{flex}	weighted reachability [3]	scope and device/grid state at fixed reach
BR_{auth}	k -zero-day safety [4]	credential, session, and command time
BR_{phys}	impact metric on reach	exogenous state u and feeder physics F

ity, active-session and command persistence, state-dependent device flexibility, and feeder state.

Formally, call a measurement *reachability-only* if it is a function M of the unchanged reachable-node set $ReachSet(x)$ alone. Since $M(ReachSet(x))$ is single-valued in $ReachSet(x)$, any two configurations that share $ReachSet(x)$ but differ in an impact dimension refute the adequacy of every reachability-only M . We give three such separations. Each is accompanied by a constructive counterexample that ships as a regression test in the released analysis engine, so the claims are checkable rather than rhetorical.

a) *Proposition 1 (scope separation).*: There exist policies π_1, π_2 and a compromise x with $ReachSet(x, \pi_1) = ReachSet(x, \pi_2)$ but $BR_{flex}(\pi_1) \neq BR_{flex}(\pi_2)$. *Construction*: hold the reachable DER set fixed and vary only the scope map A , granting control function sets under π_2 and read-only status under π_1 at the same nodes. The feasible command sets, and hence the flexibility intervals, differ. *Interpretation*: scope, not reach, sets how large a lever the adversary holds.

b) *Proposition 2 (temporal separation).*: There exist credential and session policies with equal initial reach but different retained-authority trajectories, hence different BR_{auth} and BR_{exp} . *Construction*: an ephemeral attestation-gated credential loses acceptance at the next un-attested renewal, while a long-lived unrevoked credential retains it; identical reach, different $\int BR_{flex} 1[effective] dt$. A static snapshot has no time coordinate to express this. *Interpretation*: persistence is a property of the graph's trajectory over time, not of a single snapshot.

c) *Proposition 3 (physical-state separation).*: There exist feeder states u_1, u_2 with identical cyber authorization but different physical outcomes, hence $BR_{phys}(u_1) \neq BR_{phys}(u_2)$ at fixed x and π . *Construction*: the same authorized curtailment relieves an overvoltage at one operating state and drives a reverse-power protection trip at another. No privilege graph contains u or F . *Interpretation*: realized consequence depends on feeder physics and operating conditions that are outside any authorization model.

Table III summarizes, for each dimension, the closest classical metric and what a reachability-only reading omits. The three classical metrics are static-snapshot and cyber-only.

d) *Worked counterexample.*: Two compromises reach an identical DER set D : one 20MW battery and ten rooftop PV under a common aggregator. Their reachable sets are identical, as is every topological reach metric. They differ

TABLE IV
TWO COMPROMISES WITH IDENTICAL TOPOLOGICAL REACH.

	α (attested, ephemeral)	β (legacy, long-lived)
Reachable set	D (identical)	D (identical)
Scope A_π	volt-var on PV; read-only battery	full DERControl incl. battery
Lifetime	6 h, attestation-gated	indefinite; no revocation
BR_{flex}	tens of kVAr	tens of MW
BR_{exp}	bounded (hours)	up to the cert lifetime
BR_{phys} at a vulnerable u	negligible	protection trip / excursion

only in policy and grid state (Table IV). Extending a classical graph to distinguish them requires resource-by-scope product nodes, a time index, and attached u and F ; that extension is precisely the policy-and-state model of Section III, so it agrees with our claim rather than refuting it, and such extended models are complementary to, not the same as, reachability-only measurement.

We use the propositions to motivate the model, not to dominate the paper: the contribution is the containment system and its measured evaluation, and the separations say which quantities that evaluation must move.

V. CONTAINER DESIGN

CONTAINER separates the layers that IEEE 2030.5 conflates and enforces credential lifetime at the session and command layers, not only at issuance. The design is a containment chain: fresh attestation, a fresh hardware-bound key, a short scoped operational credential, a bounded session, and a bounded command effect, so that removing authority actually removes control. Each link targets a dimension of the model (Fig. 1).

A. Components and trust layering

The device keeps a long-lived, hardware-rooted bootstrap identity [9] used only for enrollment and attestation, never accepted directly for DER control. An *attester* produces fresh evidence bound to a freshly generated operational public key; a *verifier* checks evidence against endorsement and reference values and returns an explicit result; a *credential issuer* signs a short-lived operational certificate embedding role, site, scope, and expiry only for a valid attestation result; an *authorization service* maps the operational identity to least-privilege function-set assignments; the *2030.5 server or proxy* authenticates the operational certificate, enforces a maximum session age, and rechecks authorization per command or at a bounded interval; a *session controller* and a *command-lifecycle controller* bound authority beyond expiry; and a *risk controller* selects deny, scope narrowing, or bounded emergency renewal. We use the Entity Attestation Token [10] under the RATS Attester/Verifier/Relying-Party model [11], and define a concrete EAT profile rather than treating EAT as a complete attestation protocol.

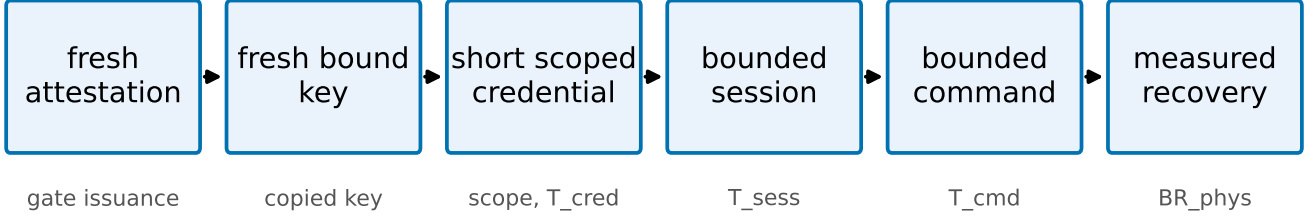


Fig. 1. The CONTAINER containment chain. Each link bounds a dimension: attestation gates issuance, fresh-key binding contains a copied key, scope and short lifetime bound BR_{flex} and T_{cred} , session and command enforcement bound T_{sess} and T_{cmd} , and measured recovery bounds BR_{phys} .

B. Fresh-key binding

Attesting a device is not sufficient: a healthy device can pass attestation while an attacker retains a previously copied operational key. CONTAINER therefore requires that every credential epoch use a newly generated operational key, generated in a TPM, TEE, or secure element where available and marked non-exportable, whose public-key digest is carried in the attestation evidence. The issuer signs only the attested public key, refuses renewal that presents only a prior certificate, rejects replayed evidence and reused nonces, and rotates the key even when the device stays healthy. This is what contains a copied key across the next epoch.

C. Session and command-effect enforcement

Certificate expiry does not end an open TLS session or an already-issued long-duration control. CONTAINER closes sessions at credential expiry (or bounds session age to the credential TTL) and revalidates authorization per command, and it bounds adversary-issued controls through a maximum command duration, cancellation on containment, and safe-default restoration, with explicit handling of scheduled DER-Controls. The two quantities the prototype measures are

$$\Delta_{expiry} = t_{last\ adv.\ command} - t_{expiry}, \quad (7)$$

$$\Delta_{effect} = t_{effect\ ends} - t_{expiry}, \quad (8)$$

each targeted to zero or a small declared bound. These are exactly the T_{sess} and T_{cmd} components of BR_{auth} .

D. Safe-mode policy

Failure behavior is explicit rather than fail-open. The risk controller chooses among the states in Table V; no failure mode grants indefinite authority, and the safety of each degraded state is validated under feeder conditions rather than assumed.

E. Migration compatibility

We separate three compatibility levels and test each rather than asserting “backward compatible.” *Protocol compatibility*: 2030.5 messages remain standard conformant. *Implementation compatibility*: unmodified client or server software can participate. *Deployment compatibility*: legacy and upgraded devices

TABLE V
SAFE-MODE AUTHORIZATION STATES.

State	Allowed	Denied
Healthy	normal scoped control	out-of-scope control
Verification delayed	read telemetry, hold last safe schedule	new high-impact dispatch
Attestation uncertain	read-only, bounded emergency control	arbitrary set-points
Known compromise	read-only or total denial	all control
Issuer outage	prior low-risk scope, bounded grace	scope expansion, high impact

coexist. Three tiers realize this: a bootstrap-only legacy tier, a gateway-attested proxy tier where an edge gateway holds the operational credential for downstream legacy devices, and a native device-attested tier. We claim only the compatibility level demonstrated by an interoperability test; if only the proxy tier interoperates with an unmodified endpoint, we say CONTAINER supports incremental deployment through a credential-terminating gateway proxy, and no more.

F. Implementation status

The design is realized as an issuance and renewal service using real X.509 certificates and mutual TLS, with a Python IEEE 2030.5-style client in place of the EPRI C client and structured event logging for every credential, session, and command transition. Its security invariants and workstation overhead are reported in Section IX; a native EPRI-client integration and constrained-hardware measurements remain future work.

VI. CYBER-SIDE ANALYSIS ENGINE

We implement the cyber-side radii as an analysis engine, `pkimodel`, that computes BR_{reach} , BR_{flex} , and BR_{auth} for a compromise locus under a policy. The engine models the identity and delegation graph over the six node types of the threat model, with typed issuance, delegation, and authorization edges. Its central modeling choice follows the formalism: a credential’s reach (which DERs it can address) is represented independently of its scope (which function sets it may invoke at each DER). This is what makes the

separating counterexample expressible in code, and we ship that counterexample as a regression test: the two compromises resolve to an identical reachable set of eleven DERs, yet the engine computes zero commandable kW for the attested credential against tens of MW for the legacy one, and a persistence of hours against the analysis horizon.

The authorization policy is parameterized, not fixed. A hard-coded permissive access model would overstate what a stolen certificate can command, so the engine exposes four ACL realism levels spanning single-device, site, feeder-segment, and whole-aggregator-fleet scope, and these are varied in the evaluation. On a synthetic 200-device fully-central fleet, the reachable-DER count grows from one to one to fifty to two hundred across the four levels, and commandable capacity grows in step. This decomposition operationalizes RQ1: most of the reach gained by a compromise is attributable to authorization scope rather than to authentication alone. The retained-authority dimension is returned as a distribution over duration rather than a point estimate, reflecting the stochastic outcome of attestation-gated renewal, and it separates credential, session, and command-effect components rather than reporting certificate lifetime alone.

The engine consumes synthetic scenarios from a deterministic, seedable generator, since real utility authorization graphs are not public; the generator is released as part of the artifact. The engine is cyber-side only. It produces the reachable-DER set and the commandable-capacity envelope that the feeder co-simulation consumes to compute BR_{phys} .

VII. FEEDER CO-SIMULATION METHODOLOGY

The physical consequence radius is measured by a feeder co-simulation that translates an authorized-control reach from the analysis engine into a malicious dispatch program and runs it on a public distribution feeder. We take the IEEE 8500-node feeder as the primary benchmark and the PNNL 9500-node system as an advanced validation tier. Time-series inputs come from public solar and load datasets. This part of the work is methodologically routine relative to the false-data-injection literature on distribution systems, so we build it to be unimpeachable rather than novel.

Two choices harden it against the standard objections to simulation studies. First, we cross-check OpenDSS against an independent GridLAB-D solver on the base case and on at least one attack case, following the precedent of the PNNL 9500 report, and we report any divergence rather than suppressing it. Second, credential state gates dispatch inside the co-simulation through a HELICS federation, so that a denied or narrowed credential actually prevents a command in simulation rather than being applied after the fact. The attack families are volt-var curve distortion, volt-watt curve distortion, export-limit distortion, and curtailment spam. The metrics are the count of nodes violating the ANSI C84.1 bands, maximum voltage deviation, the area under the violation curve over time, regulator tap operations, unnecessary curtailment, energy not served, and recovery time to steady limits, with the per-unit voltage excursion as the primary continuous measure and a protection-trip flag as the discrete outcome.

We do not tune feeder parameters to amplify impact; we report what the standard models produce. The clean base case, the four attack families, the eight metrics, and the cross-solver agreement are the acceptance criteria for this component. As with the credential service, a first benchmark run on the IEEE 8500-node feeder appears in the preliminary evaluation; the full sweep on both feeders, with the GridLAB-D cross-check and seasonal operating states, follows the pre-registered protocol chosen in advance.

VIII. EVALUATION PROTOCOL (PRE-REGISTERED)

No measured results are reported in this section. We state the evaluation protocol, which is frozen and content-hashed before any run and deposited on a trusted timestamped repository, so that the study design cannot be adjusted after seeing outcomes and so that a private hash is not the only immutability evidence. Chronologically, the measured results in Section IX are pilot experiments; the confirmatory scenario matrix and statistical plan were frozen and hashed after the pilots, and pilot observations are excluded from the confirmatory tests.

a) Baselines and mechanism ablations.: Six credential-lifecycle designs are compared: legacy identity-as-authority (B1), legacy with a narrow access list (B2), ephemeral-only (B3), attestation-only (B4), full CONTAINER (B5), and safe-mode renewal (B6). Each baseline is specified by explicit fresh-key, session-enforcement, and command-cleanup attributes, because the central claim is that short lifetime alone does not contain a copied key, an active session, or a persistent command. Eight mechanism ablations (A1–A8) each remove one link of the containment chain, so the measured reduction is attributed to a mechanism rather than to shorter certificates. Table VI gives the baseline mechanism attributes.

b) Factors and design.: The factors are credential time to live, authorization scope, compromise locus (with the issuer treated as a trust boundary, not a defended case), DER penetration and technology mix, attack program, attestation-trust state, renewal policy, and operating state. Every attack runs under both a worst-case optimizing attacker and an operationally realistic telemetry-limited attacker, so no result is only an omniscient upper bound. The primary comparison is the fully-crossed lifetime-by-scope factorial, whose interaction term is the compositional claim.

c) Statistics.: The experimental unit is the complete paired scenario run, not the per-timestep sample. The primary voltage-violation area is analyzed with a mixed-effects model, $\log(1 + Y) = \beta_0 + \beta_1 \log(\text{TTL}) + \beta_2 \text{Scope} + \beta_3 \log(\text{TTL}) \times \text{Scope} + b_{\text{day}} + b_{\text{seed}} + \epsilon$, with the interaction coefficient testing the primary hypothesis and robust, aligned-rank, or permutation fallbacks where assumptions fail. Protection trips use mixed-effects logistic regression, violating-node counts a negative-binomial mixed model, and recovery time survival analysis. Effect sizes are the median paired difference, the ratio of geometric means, the paired rank-biserial correlation, and odds ratios, each with bootstrap confidence intervals, and Holm correction is applied within, not across, hypothesis families. A five-seed pilot and a simulation-based power analysis

TABLE VI
CREDENTIAL-LIFECYCLE BASELINES. FK = FRESH KEY PER EPOCH; SESS = SESSION ENFORCEMENT; CMD = COMMAND-EFFECT CLEANUP.

ID	Design	Lifetime	Attestation	Scope	FK	Sess	Cmd
B1	legacy identity-as-authority	long	no	deployment ACL	no	no	no
B2	legacy + narrow ACL	long	no	narrow	no	no	no
B3	ephemeral-only	short	no	reference	opt.	expiry	opt.
B4	attestation-only	long	yes	reference	enroll	weak	no
B5	full CONTAINER	short	yes	narrow	per-epoch	yes	yes
B6	CONTAINER safe-mode	short	yes	narrow/degraded	per-epoch	yes	yes

set the production seed count before final comparisons are viewed.

d) *Pre-registered hypotheses and honesty rule.*: The primary hypothesis is a non-additive lifetime-by-scope interaction on voltage-violation area. Secondary hypotheses cover capacity-time exposure reduction, the advantage of the full design over both the ephemeral-only and attestation-only ablations, the reduction of post-expiry residual authority by session and command enforcement, locus dominance, the safe-mode availability tradeoff, fresh-key containment of a copied key, and the renewal-latency safety margin. The full design must reduce capacity-time exposure and voltage-violation area on at least one public feeder, which is the study's go/no-go gate. The expected-pattern statement records what we anticipate; any deviation, including a weak physical effect on a robust feeder, is reported rather than engineered away, and severe or nonconverging runs are never silently dropped.

IX. PRELIMINARY EVALUATION

We report a preliminary evaluation that exercises the full pipeline end to end and produces measured results. Its scope is deliberately limited and stated plainly: a workstation platform (no constrained hardware), the real IEEE 8500-node feeder as a first benchmark run but not yet the PNNL 9500 system, a single solver (OpenDSS, no GridLAB-D cross-check), load-state snapshots rather than a full seasonal time series, one PV penetration level, a fixed rather than per-state-optimized attacker, and a Python IEEE 2030.5-style client in place of the EPRI C client. These results establish that the mechanism and the measurement work; the pre-registered full-scale sweep is not replaced by them. Table VII states the status of each component.

a) *Functional invariants and containment (C1, C2).*: The credential service uses real X.509 / ECDSA certificates and an EAT-style attestation token. Ten implemented security invariants all pass: bootstrap identity cannot authorize control, issuance requires a valid fresh attestation, the certificate is bound to a freshly generated operational key, replayed evidence is rejected, an expired credential cannot issue commands, per-command scope matches issuance, a bad measurement yields only a narrowed safe-mode scope, a prior epoch cannot self-renew, and a copied prior operational key is contained at the next epoch. With per-command revalidation the post-expiry acceptance gap is $\Delta_{\text{expiry}} = -0.5$ s (the last accepted adversarial command precedes expiry), against +5.0 s without

TABLE VII
IMPLEMENTATION AND EVALUATION STATUS.

Component	Status
Cyber-side analysis engine	implemented, measured
Credential service (issuance, fresh key, session/command enforcement)	implemented, measured
Feeder co-simulation (OpenDSS, two feeders)	implemented, measured
Availability fault injection	implemented, measured
Native EPRI-client 2030.5 traffic	planned
Constrained-hardware / TEE attestation	planned
GridLAB-D cross-solver check	planned
Interoperability tests	planned
Full pre-registered sweep	planned

TABLE VIII
SECURITY INVARIANTS (ALL PASS ON THE PROTOTYPE, REAL X.509/MTLS).

ID	Invariant	Result
I1	Bootstrap identity cannot authorize control	pass
I2	Issuance requires a fresh attestation	pass
I3	Credential bound to a fresh public key	pass
I4	Replayed evidence rejected	pass
I5	Expired credential cannot issue a command	pass
I6	Scope enforced per command	pass
I7	Bad measurement yields degraded/denied scope	pass
I8	Prior epoch cannot self-renew	pass
I9	Copied prior key fails after rotation	pass
I10	Active session/command bounded past expiry	pass

enforcement, where it grows to the cached session age. A fresh operational key is generated each epoch, so a copied key is useless after rotation. Table VIII lists the invariants and their outcomes.

b) *Overhead (C3).*: On the workstation tier, over 500 iterations (100 for the handshake), the median software-path latencies (no hardware root of trust, no network transport) are 0.015 ms for key generation, 0.05 ms for attestation construction, 0.2 ms for a full attested issuance, and 1.1 ms for a real mutual-TLS handshake, with p99 tails within a few milliseconds; the issuer sustains about 4900 issuances per second single-threaded. Renewal is therefore inexpensive on this platform. The measured operations are standard EC P-

256 primitives dominated by a single ECDSA signature and verification, so per-issuance cost on a constrained inverter controller can be estimated from its public-key throughput; a hardware-backed measurement is declared future work rather than claimed here.

c) *Availability (C6).*: Renewal is a full attested issuance, so its cost is the issuance cost above; the renewal safety margin $T_{\text{TTL}} - (T_{\text{renew,p99}} + T_{\text{net}})$ stays positive at every tested lifetime down to five minutes, and a 100,000-device fleet re-mints in about 20 s single-threaded, so short lifetimes are latency-feasible. The availability risk is instead a verifier or issuer outage longer than the credential lifetime. Injecting verifier outages into the running service (renewal fails during the outage), fail-closed denial retains 0.77 of legitimate control availability over the outage window, a bounded grace period 0.83, and degraded-scope safe mode 1.0; control recovers as soon as the verifier returns. The injection is in-process; a networked or hardware outage, clock skew, and renewal storms remain for the full study.

d) *Feeder consequence (C5).*: We run an export/overvoltage attack on the real IEEE 8500-node feeder (4876 buses, 1177 loads, 12 MW) with 600 seeded rooftop-PV units, ten paired seeds, under three load states. Table X reports the attack-induced overvoltage-violation area ΔJ_V , the median over seeds against the base case at the same load. Because the 8500 base has native undervoltage at its extremities, we report the induced *overvoltage* area the attack targets rather than a two-sided area that would credit the attack for lifting sagging nodes; this attack-specific endpoint was chosen after observing the base undervoltage, while the two-sided area remains the declared primary system-level endpoint. Under light load and high PV a legacy full-authorization compromise induces a median $\Delta J_V = 126.6$ (range 108–134); the CONTAINER narrowed scope keeps it within the ANSI band (median $\Delta J_V \leq 0$). Under normal load the figures are 6.6 against 0.1. Under heavy load the export attack cannot raise voltage above the band, so $\Delta J_V \approx 0$ regardless of policy. The ephemeral-only baseline equals the legacy one physically, since a short lifetime does not narrow scope; its advantage is temporal. The consequence is strongly state dependent, the physical-state separation of Proposition 3 on the real benchmark.

e) *Time-series containment and mechanism decomposition (C5).*: To measure temporal containment directly, rather than as a product of a measured and a modeled factor, we run each policy as one time-series: a compromised credential drives the DER from minute five, and its lifecycle then determines whether the malicious setpoint keeps affecting the feeder. We integrate the real induced overvoltage area over time, in p.u.-node-minutes (Fig. 2). A legacy credential (B1) holds the feeder in violation for the whole horizon (integral 6956, no recovery). Holding the broad scope fixed but adding the CONTAINER lifecycle (A4) halves the integral to 3196 and the feeder recovers at credential expiry, which isolates the temporal effect of lifetime with session and command enforcement at equal scope. Removing session enforcement (A2) or command cleanup (A3) from that configuration returns the feeder to no recovery (7030), so each mechanism is necessary.

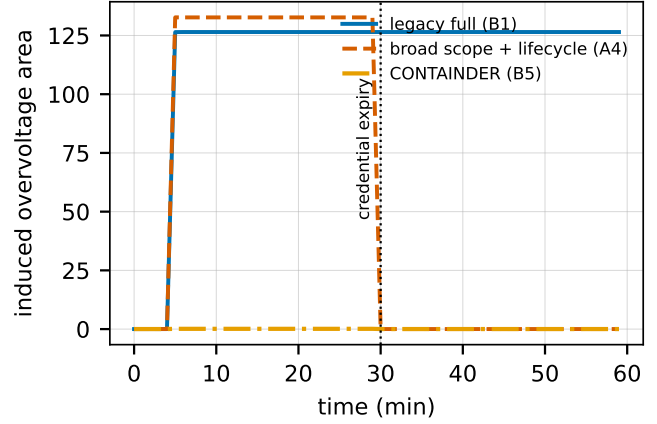


Fig. 2. Time-series induced overvoltage on the IEEE 8500-node feeder (light load, one seed). A legacy credential (B1) never recovers; at the same broad scope the CONTAINER lifecycle (A4) recovers at credential expiry; narrow scope (B5) additionally bounds the instantaneous envelope. The session-off and command-cleanup-ablations track B1 and are omitted for clarity.

Narrowing scope (B5) additionally bounds the instantaneous envelope, and the integral falls to 3.8. This single experiment gives the equal-scope decomposition the design intends: scope bounds the envelope, lifetime with enforcement bounds the duration, and neither alone suffices.

f) *Policy comparison and the persistence-scope interaction (C5, H1).*: As a first-order summary that separates all six baselines at once, we also report a proxy combining the measured physical envelope with the modeled retained authority, $\Delta J_V \times \text{BR}_{\text{auth}}$ (Table IX); this is a physical-impact-times-duration quantity, not the flexibility-time BR_{exp} integral of the model, and the time-series above is the direct measurement it approximates. At full authorization scope, lengthening persistence from the ephemeral B3 (retained authority ≈ 10 h) to the legacy B1 (≈ 2 years) multiplies light-load exposure roughly 1700 $\times$, from 1.3×10^3 to 2.2×10^6 . At narrow scope the same change is inert: B2 and B5 are both near zero. The effect of persistence therefore depends on scope, consistent with the pre-registered persistence-by-scope hypothesis; because exposure is a product of a measured and a modeled factor, this preliminary pattern does not by itself establish the pre-registered mixed-effects test over the full sweep. The full design B5 has lower exposure than both the ephemeral-only B3 and the attestation-only B4 (8.3×10^3), so neither half alone suffices. Mechanism ablations attribute the reduction: removing scope narrowing restores physical exposure (from near zero to 1.5×10^3), while fail-open renewal inflates retained authority from 12 h to 1.75×10^4 h. Scope narrowing bounds the physical envelope; attestation-gated renewal with session and command enforcement bounds how long it persists, the temporal containment the C2 copied-key and session-theft results measure.

g) *Penetration and robustness.*: Induced overvoltage grows with PV penetration under a legacy full-authorization compromise, from 8 at 20% to 254 at 100%, while the CONTAINER narrowed scope holds it near zero at every level (Fig. 4). The same containment holds across attack

TABLE IX
FIRST-ORDER EXPOSURE PROXY $\Delta J_V \times BR_{auth}$ BY BASELINE (LIGHT LOAD, IEEE 8500, N=10; ΔJ_V MEASURED, BR_{auth} MODELED). THE TIME-SERIES (FIG. 2) IS THE DIRECT MEASUREMENT.

Baseline	Scope	BR_{auth}	Exposure
B1 legacy full	full	~ 2 yr	2.2×10^6
B2 ACL-narrow	narrow	~ 2 yr	≈ 0
B3 ephemeral-only	full	~ 10 h	1.3×10^3
B4 attestation-only	reference	~ 14 d	8.3×10^3
B5 full CONTAINER	narrow	~ 12 h	≈ 0
B6 safe-mode	narrow	~ 12 h	≈ 0

TABLE X
ATTACK-INDUCED OVERVOLTAGE-VIOLATION AREA ΔJ_V ON THE IEEE 8500-NODE FEEDER (MEDIAN OVER 10 SEEDS).

Operating state	Legacy full (B1)	CONTAINER (B5)
light load / high PV	126.6	-0.01
normal	6.6	0.13
heavy load	≈ 0	≈ 0

families (Fig. 5), and capacity-time exposure separates all six baselines along the persistence–scope interaction (Fig. 3). It also holds on a second public feeder: on the IEEE 123-bus system a legacy full-authorization compromise induces overvoltage under high PV penetration (induced area up to 2.2) that CONTAINER holds at zero, though the stiffer regulated 4.16 kV feeder requires higher penetration than the 8500 to breach the band, consistent with the feeder-dependence of Proposition 3.

h) Attack families.: The containment holds across attack families, not only the export attack (five seeds, light load). A legacy full-authorization compromise induces overvoltage areas of 122 (export-limit), 97 (volt-var, reactive injection alone), and 57 (volt-watt, active injection) that the CONTAINER narrowed scope reduces to zero in every case; for a curtailment-spam attack, narrowing cuts the per-node voltage variability from 0.012 to 0.004 p.u., a factor of three. Overvoltage is reachable through active or reactive authority, and narrowing scope removes both.

i) Least privilege preserves legitimate utility (C5).: A scope that removed all control would contain the attack trivially, by disabling the device. CONTAINER’s operational scope is bounded volt-var support, not read-only. Under it a legitimate reactive voltage-support command is authorized while an active-power export command is denied; the attacker’s only remaining lever is bounded reactive, which induces overvoltage 0.04 against 126 under full authorization, whereas a read-only scope authorizes neither the legitimate nor the malicious command. The narrow scope therefore retains the most common legitimate DER function while bounding the attack, which distinguishes least privilege from denial of service.

j) Limitations.: These results are preliminary and we do not generalize them. Physical results come from two public feeders (a ten-seed IEEE 8500 sweep and a confirmatory IEEE 123 run) and a single solver, with a fixed export attacker; the

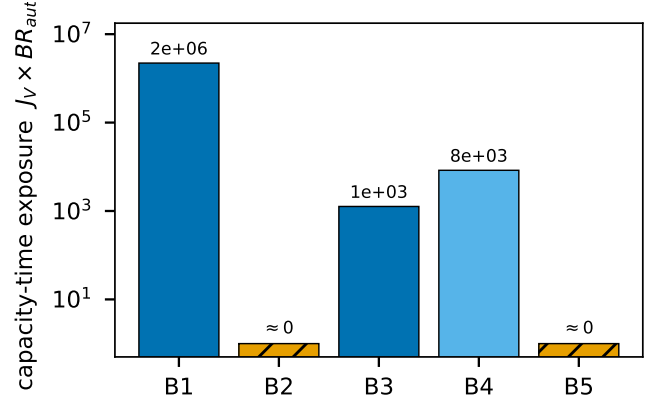


Fig. 3. Capacity-time exposure by baseline on the IEEE 8500-node feeder (light load, log scale). Persistence multiplies exposure only at full scope (B1 vs B3); at narrow scope (B2, B5) it is inert, the persistence–scope interaction. Bars floored at 1 for log display; B2, B5, B6 are ≈ 0 .

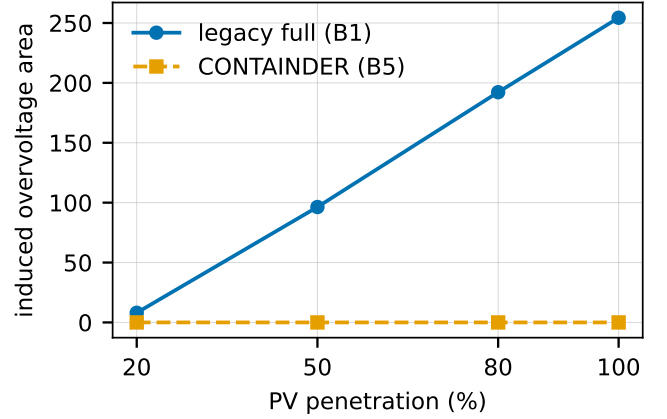


Fig. 4. Induced overvoltage area versus PV penetration (IEEE 8500, light load, 5 seeds). Harm scales with penetration under the legacy baseline; containment is penetration-independent under CONTAINER.

IEEE 9500-node system was obtained but did not converge in our OpenDSS build and is deferred to an appendix tier per the study’s pre-registered contingency. Retained-authority figures are model outputs under assumed attestation and lifetime parameters, not measured field values, and the exposure metric inherits those assumptions. The availability results model the outage response rather than injecting real verifier outages on hardware. Incremental deployment is by construction; interoperability against an unmodified endpoint is untested. The pre-registered sweep on two public feeders, with a cross-solver check, seasonal states, constrained hardware, and the full seed budget, is what would support quantitative claims; the 8500-node results are the primary feeder evidence here.

X. RELATED WORK

We position CONTAINER against ten strands (Table XI). *IEEE 2030.5 and CSIP security.* The 2018 and 2023 editions of IEEE 2030.5 mandate mutual TLS with X.509 device certificates [6], [7], constrained for interconnection by the Common Smart Inverter Profile [8]. The closest prior work is

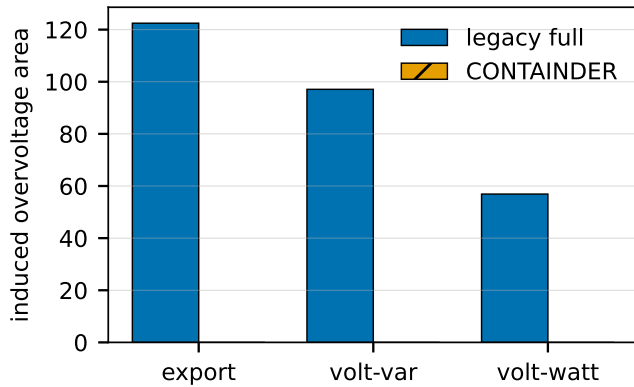


Fig. 5. Induced overvoltage area by attack family (IEEE 8500, light load). Narrowing scope contains overvoltage driven by active or reactive authority.

the 2030.5 security tutorial of Passos et al. [5], a survey that proposes no redesign, quantifies no feeder consequence, and formalizes no impact metric; Qi et al. survey the broader DER and smart-inverter surface [12]. The Sandia analysis supplies the revocation constraints we respect [1].

DER and smart-inverter attacks. High-wattage IoT botnets can move the grid [13], [14]; on the DER side, inverter attacks destabilize grid and market [15] and distribution voltage control is attackable under high PV [16], while the SUN:DOWN disclosure catalogs fleet-scale inverter vulnerabilities [17]. We tie the attack surface to the credential lifecycle rather than assuming device compromise.

Distribution-grid cyber-physical impact. False-data-injection work quantifies compromise consequence [18]–[20], grounded by the 2015 Ukraine incident [21] and quantitative risk formulations [22]; our feeder method is supporting, not novel.

Short-lived certificates and PKI lifecycle. ACME [23] and Let’s Encrypt [24] automate issuance, SCEP [25] is the incumbent device enrollment, and HTTPS-ecosystem measurement quantifies the lifecycle problem [26]; we adopt short lifetimes as one link of a chain.

Hardware-bound identity. Secure device identity [9], firmware TPM [27], and hardware DICE [28] root our bootstrap layer.

Remote attestation. We use the RATS architecture [11], the Entity Attestation Token [10], and BRSKI [29], on established principles [30], adding fresh-key binding.

Attack-graph and cyber-physical metrics. Logic-based [2], [31], [32], probabilistic and Bayesian [3], [33], [34], and k -zero-day [4] metrics are static-snapshot and cyber-only; Section IV shows what reachability-only reading omits, positioning our model as complementary.

Co-simulation. HELICS federates the cyber and grid models [35], [36] with GridLAB-D on the power side [37].

DER authorization. IEEE 1547 [38] defines the DER functions, DERMS perform feeder voltage management [39], and role-based access control [40] abstracts the scoping we narrow and enforce.

Revocation at scale. Web-PKI revocation delivery is gap-

ridden [41], scalable revocation remains an open problem [42], [43], and even OCSP Must-Staple adoption is weak [44], which motivates containment by short-lived credentials rather than by revocation.

XI. ETHICS AND RESPONSIBLE DISCLOSURE

This work studies the consequences of credential compromise in order to bound them, and it carries a corresponding responsibility. All experiments use synthetic and lab-generated certificates. We use no real vendor keys or secrets, we test against no production DER fleet or any internet-reachable device, and we build no exploit tooling bound to a specific vendor product. The malicious dispatch programs exist only inside the feeder co-simulation and are never emitted to hardware.

The physical-consequence results describe what a compromised but protocol-conformant credential can already do under the deployed standard; they do not introduce a new vulnerability so much as measure an existing exposure that the standard’s prohibition on revocation leaves unmitigated. Where results identify a concrete deployment weakness, we will follow coordinated disclosure with the relevant standards body and affected implementers before public release, and we will time any artifact release accordingly. The redesign and its migration path are offered as the constructive counterpart to the measurement: the point of quantifying the blast radius is to shrink it.

XII. LIMITATIONS AND CONCLUSION

a) Limitations.: We state the boundaries of the present work plainly. The formal model, the separation result, and the cyber-side analysis engine are complete and tested. The attested-credential service and the feeder pipeline are implemented; a preliminary reduced-scale evaluation reports real measured results (security invariants, containment timing, workstation overhead, availability, and feeder consequence on the IEEE 8500-node feeder), but the full pre-registered sweep on two public feeders, with a cross-solver check and constrained hardware, has not yet executed and is what would support quantitative claims. The authorization graphs used by the engine are synthetic, because real utility authorization graphs are not public; we mitigate this with a released, parameterized generator and by treating access-model permissiveness as an ablated variable rather than a fixed assumption. The exact clause-level delta between the 2018 and 2023 revisions of the standard’s certificate-management text remains an open item pending access to the revised standard, and our claims about revocation rest on the primary Sandia analysis rather than on that delta.

b) Conclusion.: Credential compromise in IEEE 2030.5 is not well described by how many nodes an adversary can reach. It is described by how much physical capacity those nodes command, how long the adversary keeps it, and what the feeder does in response. We formalized these as four impact dimensions, showed that reachability-only measurement omits scope, time, and feeder state, and built a redesign that shrinks

TABLE XI
POSITIONING AGAINST REPRESENTATIVE PRIOR WORK. Y = YES, N = NO, P = PARTIAL, – = N/A.

Work	2030.5	Scope	Time	Phys.	Attest.	Proto.	Feeder
Attack-graph metrics [2]–[4]	n	p	n	n	n	–	n
2030.5 security survey [5]	y	p	n	n	n	n	n
Attestation building blocks [9]–[11]	n	p	p	p	y	p	n
CONTAINER (this work)	y	y	y	y	y	y*	y*

*Prototype and preliminary feeder results are reported in Section IX; the full sweep follows the pre-registered protocol.

them using attested, short-lived, scope-bound operational credentials with non-renewal as containment. A preliminary end-to-end evaluation is consistent with the design: the security invariants hold, per-command enforcement contains authority at expiry, and narrowing scope keeps the overvoltage a compromise induces on the IEEE 8500-node feeder within limits where a legacy credential drives it far outside, with the full evaluation still to come. The model is the lever: it says which quantities a DER-authorization defense must move, and it rejects the measures that only move reach.

DATA AND CODE AVAILABILITY

The analysis engine, the credential service, the feeder scripts, the pre-registered scenario matrix (with its recorded hash), and the result files supporting this study will be deposited in a public repository. The IEEE 8500-node model is from the public IEEE PES distribution test-feeder set.

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